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1 考点 1: 信用损失分布

1. As one of the useful approaches which can be used to estimate PD, single-factor model is growing popular. Assume there is an individual firm i , has beta correlation, β_i , with market factor, m . If the market factor is given, which of the following statements most accurately describes the implication?
- A. Individual idiosyncratic shocks are negatively correlated with other firms' shock
 - B. Individual idiosyncratic shocks are positively correlated with other firms' shock
 - C. The unconditional standard deviation is less than the conditional standard deviation
 - D. Individual asset returns are independent from other firms' shocks and returns

答案:D

解析:

- A 选项错误。**单因素模型中，个别公司的特定风险冲击与其他公司的特定风险冲击之间不存在负相关关系。特定风险冲击是相互独立的。
- B 选项错误。**单因素模型中，个别公司的特定风险冲击与其他公司的特定风险冲击之间不存在正相关关系。特定风险冲击是相互独立的。
- C 选项错误。**在单因素模型中，无条件标准差通常大于条件标准差。这是因为条件标准差只考虑了市场因素的影响，而无条件标准差还包含了特定风险的影响。
- D 选项正确。**单因素模型的一个重要含义是，个别公司的资产回报和特定风险冲击与其他公司的特定风险冲击和回报是相互独立的。这种独立性是单因素模型的基本假设之一。
- 考察:** 单因素模型中个别公司的资产回报和特定风险冲击相互独立

2. Preciousmetals is a specialist company that only trades derivatives on precious metals. Preciousmetals and a handful of other firms, all of whom have large notional outstanding contracts with Preciousmetals, dominate the market for such derivatives. Preciousmetals management would like to mitigate its overall counterparty exposure, with the goal of reducing it to almost zero. Which of the following methods, if implemented, could best achieve this goal?
- A. Ensuring that sufficient collateral is posted by counterparties
 - B. Diversifying among counterparties
 - C. Cross-product netting on a single counterparty basis
 - D. Purchasing credit derivatives, such as credit default swaps

答案:A

解析:

- A 选项正确。**从理论上讲，只要有足够数量的高质量抵押品（如现金或短期投资级政府债券），就可以几乎完全抵消交易对手的风险敞口。如果交易对手违约，持有未平仓衍生品合约的一方可以接收抵押品。
- B 选项错误。**在交易对手之间分散风险并不能将风险敞口降低到接近零。由于市场参与者数量有限，分散化效果有限，且无法完全消除风险。
- C 选项错误。**单一交易对手基础上的跨产品净额结算只能减少对其中一个交易对手的风险敞口，无法将整体风险敞口降低到接近零。
- D 选项错误。**购买信用衍生品（如信用违约互换）只是将个别交易对手的风险替换为 CDS 卖方的交易对手风险，无法将风险敞口降低到接近零。
- 考察:** 高质量抵押品可以几乎完全抵消交易对手风险敞口

3. Credit risk is a function of the probability of default, exposure at default, and loss given default. Assuming that the individual exposures at default with a counterparty are fixed, which of the following statements is correct?
- A. The probability of default can be mitigated by collateral and exposure at default can be mitigated by netting
 - B. The probability of default can be mitigated by netting and exposure at default can be mitigated by collateral
 - C. Loss given default can be mitigated by collateral and exposure at default can be mitigated by netting
 - D. Loss given default can be mitigated by netting and exposure at default can be mitigated by collateral

答案:C

解析:

A 选项错误。违约概率取决于信用事件，这些事件不能通过抵押品来控制，因为信用事件取决于支付能力和支付意愿，这两者都与抵押品无关。同时，抵押品也不能减少违约时的风险敞口。

B 选项错误。违约概率取决于信用事件，这些事件不能通过净额结算来控制，因为信用事件取决于支付能力和支付意愿，这两者都与净额结算无关。同时，抵押品也不能减少违约时的风险敞口。

C 选项正确。抵押品可以在违约后用于减少违约损失。净额结算可以在交易对手违约时减少结算金额，从而减少违约时的风险敞口。

D 选项错误。净额结算不能减少违约损失，它只能减少违约时的风险敞口。同时，抵押品也不能减少违约时的风险敞口，它只能用于减少违约损失。

考察: 抵押品减少违约损失，净额结算减少违约敞口

4.

A.

答案:B

解析:

A 选项错误。

•

B 选项错误。

•

C 选项错误。

•

D 选项错误。

•

考察:

5.

A.

答案:B

解析:

A 选项错误。

•

B 选项错误。

•

C 选项错误。

•

D 选项错误。

•

考察:

6.

A.

答案:B

解析:

A 选项错误。

-
- B 选项错误。
-
- C 选项错误。
-
- D 选项错误。
-

考察:

7.
A.

答案:B
解析:

- A 选项错误。
-
- B 选项错误。
-
- C 选项错误。
-
- D 选项错误。
-

考察:

8.
A.

答案:B
解析:

- A 选项错误。
-
- B 选项错误。
-
- C 选项错误。
-
- D 选项错误。
-

考察:

9.
A.

答案:B
解析:

- A 选项错误。

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B 选项错误。
·
C 选项错误。
·
D 选项错误。
·
考察:

10.
A.
答案:B
解析:
A 选项错误。

·
B 选项错误。
·
C 选项错误。
·
D 选项错误。
·
考察:

11.
A.
答案:B
解析:
A 选项错误。

·
B 选项错误。
·
C 选项错误。
·
D 选项错误。
·
考察:

2 考点 2: 违约相关性和信用 VaR

1. A manager of a mutual fund has taken significant credit exposure to Europe and Asia. Concerned with uncertain market conditions, the manager wants to change the assumptions in the fund’ s risk models by increasing the default correlation between bonds issued in Europe and bonds issued in Asia. If the default correlation is increased and all the other parameters are kept the same, which of the following is true?

- A. The expected loss of the portfolio will increase.
- B. The unexpected loss of the portfolio will decrease.
- C. The expected loss of the portfolio will decrease.
- D. The unexpected loss of the portfolio will increase

答案:D

解析:

A 选项错误。

•

B 选项错误。

•

C 选项错误。

•

D 选项正确。

- Default correlation has no effect on the expected losses of the portfolio, but increases the unexpected losses of the portfolio. 违约相关性对投资组合的预期损失没有影响， 但会增加投资组合的非预期损失。

考察:

2. Consider a pair of two speculative credits, rated BB and BB-, with default probabilities respectively of 2% and 3%. If their joint default probability is 0.4%, which is nearest to the implied default correlation?

- A. Zero
- B. 0.083
- C. 0.1424
- D. 0.3750

答案:C

解析:

A 选项错误。

•

B 选项错误。

•

C 选项正确。

- $\text{Default correlation} = (0.4\% - 2\% \times 3\%) / [\text{sqrt}(2\% \times 98\%) \times \text{sqrt}(3\% \times 97\%)] = 0.142365.$

D 选项错误。

•

考察:

3. Which of the following statements best describes the calculation of implied correlation?

- A. The implied correlation for the mezzanine tranche assumes non-constant pairwise correlation.

- B. Observable market prices of credit default swaps are used to infer the tranche values.
- C. The tranche pricing function is calibrated to match the model price with the market price.
- D. The risk-adjusted default probabilities are used in model calibration.

答案:C

解析:

A 选项错误。

•

B 选项错误。

•

C 选项正确。

- Starting with observed market prices and a pricing function for the tranches, it is possible to back out the implied correlation to calibrate the model price with the market price. The computation of implied correlation assumes constant pairwise correlation. Both credit default swap and tranche values are observed. Observed tranche values are used in conjunction with risk-neutral default probabilities to compute implied correlation. 隐含相关系数的计算假设两两相关系数不变。信用违约互换 (CDS) 和部分层级的价值可以被观察到。观察到的层级价值与风险中性的违约概率结合起来计算隐含相关性

D 选项错误。

•

考察:

4. Consider two portfolios. One with USD 100 million credit exposure to a single B-rated counterparty. The second with USD 100 million on credit exposure split evenly between 50 B-rated counterparties. Assume that default probabilities and recovery rates are the same for all B-rated counterparties. Which of the following is correct?

- A. The expected loss of the first portfolio is greater than the expected loss of the second portfolio AND the unexpected loss of the first portfolio is greater than the unexpected loss of the second portfolio.
- B. The expected loss of the first portfolio is greater than the expected loss of the second portfolio AND the unexpected loss of the first portfolio is equal to the unexpected loss of the second portfolio.
- C. The expected loss of the first portfolio is equal to the expected loss of the second portfolio AND the unexpected loss of the first portfolio is equal to the unexpected loss of the second portfolio.
- D. The expected loss of the first portfolio is equal to the expected loss of the second portfolio AND the unexpected loss of the first portfolio is greater than the unexpected loss of the second portfolio

答案:D

解析:

A 选项错误。

•

B 选项错误。

•

C 选项错误。

•

D 选项正确。

- There’ s diversification effect in unexpected loss 非预期损失存在分散化效应

考察:

5. Suppose a credit position has a correlation to the market factor of 0.5. What is the realized market value that is used to compute the probability of reaching a default threshold at the 99% confidence level?

- A. -0.2500.
- B. -0.4356.
- C. -0.5825.
- D. -0.6243.

答案:D

解析:

A 选项错误。

•

B 选项错误。

•

C 选项错误。

•

D 选项错误。

- A default loss level of 0.01 corresponds to -2.33 on the standard normal distribution. The realized market value is computed as follows:

考察:

6. The single-factor model is used to examine the impact of varying default correlations based on a credit position’ s beta. Each individual firm or credit, i , has a beta correlation, β_i , with the market, m . Which of the following statements most accurately describes the implication of using a specific value m for the market parameter in the single-factor model?

- A. The conditional probability of default will be greater than the unconditional probability of
- B. The unconditional standard deviation is less than the conditional standard deviation.
- C. Individual idiosyncratic shocks, ϵ_i , are positively correlated to other firms’ shocks.
- D. Individual asset returns, a_j , are independent from other firms’ shocks and returns.

答案:D

解析:

A 选项错误。

•

B 选项错误。

•

C 选项错误。

•

D 选项正确。

- Important implication of the single-factor model is that individual asset returns, a_i , and idiosyncratic shocks, ϵ_i are independent from other firms’ shocks and returns.

考察:

7. A risk analyst at a bank is reviewing the annex to a report on credit risk models that the risk department has prepared for the bank’ s risk committee. The analyst takes note of the features of the KMV model and the CreditMetrics model as well as the benefits and drawbacks of using the two models. Which of the following statements about these models is correct?

- A. In the CreditMetrics model, the value of a firm can change continually but the firm’ s default probability may remain the same if there is no change in the firm’ s rating.

- B. A shortcoming of the KMV model is that it cannot be used to compute the risk of a liabilities portfolio that includes both a zero-coupon debt and a convertible debt.
- C. The CreditMetrics model assumes a simple capital structure of the firm, which renders no need to estimate correlations among the different bonds in the liabilities portfolio.
- D. The KMV model relies on the mark-to-market value of debt while the CreditMetrics model applies a factor model which assumes the risk factors follow a gamma distribution.

答案:A

解析:

A 选项正确。

- CreditMetrics 模型基于信用评级的变化来评估信用风险。在该模型中，违约概率（PD）主要依赖于公司的信用评级。如果评级没有发生变化，即使公司价值波动，违约概率也可能保持不变。因此，这一描述符合 CreditMetrics 模型的核心逻辑。

B 选项错误。

- KMV 模型利用 Merton 模型的思想，通过股权市场价值和波动率推导出公司资产价值及其波动性，从而估算违约距离（Distance to Default, DD）和违约概率。KMV 模型能够处理包括零息债券和可转换债券在内的多种债务工具。因此，认为 KMV 模型无法计算包含零息债券和可转换债券的投资组合风险的说法是错误的。

C 选项错误。

- CreditMetrics 模型并不假设公司资本结构简单。相反，它考虑了不同债券之间的相关性，并使用因子模型（如高斯 Copula 模型）来模拟这些相关性。因此，认为该模型无需估计债券间相关性的说法是不正确的。

D 选项错误。

- KMV 模型确实依赖债务的市场价值（Mark-to-Market Value），但 CreditMetrics 模型中的因子模型假设风险因子服从正态分布（Normal Distribution），而非伽马分布（Gamma Distribution）。因此，这一描述不符合事实。

考察:

8. A senior risk manager at an investment bank has been asked to give a presentation on correlation risk to a group of junior risk analysts. The manager gives several examples to demonstrate the impact of correlation risk to market and credit risks and how this impact is measured. Which of the following statements would be correct to include in the manager’s presentation?

- A. A. VaR explicitly incorporates correlation risk into risk calculations through a multiplier based on the measured correlation of assets in the portfolio.
- B. In assessing the risk in a typical loan portfolio, default correlations between different market sectors are higher than correlations within sectors.
- C. Investors who purchase CDS protection on a specific bond are exposed to the risk of changes in the correlation between the bond and the CDS counterparty.
- D. A creditor with a large number of debtors benefits from a high default correlation of its debtors due to an increased ability to accurately measure credit risk of default.

答案:C

解析:

A 选项错误。

-

B 选项错误。

-

C 选项正确。

-

D 选项错误。

-

考察:

9.

A.

答案:B

解析:

A 选项错误。

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B 选项错误。

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C 选项错误。

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D 选项错误。

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考察:

10.

A.

答案:B

解析:

A 选项错误。

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B 选项错误。

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C 选项错误。

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D 选项错误。

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考察:

3 考点 3: 经济资本

1. Suppose a portfolio has a value of \$1,000,000 with 50 independent credit positions. Each of the credits has a default probability of 2% and a recovery rate of 0%. The credit portfolio has a default correlation equal to 0. The number of defaults is binomially distributed and the 95th percentile of the number of defaults is 3. What is the credit value at risk at the 95confidence level for this credit portfolio?

- A. \$20,000.
- B. \$40,000.
- C. \$60,000.
- D. \$980,000.

答案:B

解析:

A 选项错误。

•

B 选项正确。

- The loss given default is \$60,000 [3 x (\$1,000,000 / 50)]. The expected loss is equal to the portfolio value times π and is \$20,000(0.02 x\$1,000,000). The credit VaR is defined as the quantile of the credit loss less the expected loss of the portfolio. At the 95% confidence level, the credit VaR is equal to \$40,000 (\$60,000 minus the expected loss of \$20,000).60000-20000=40000

C 选项错误。

•

D 选项错误。

•

考察:

2. A senior risk analyst at a bank asks an associate to calculate the unexpected loss (UL) of a single loan to a corporate client. The associate gathers the following information on the loan over a 1-year period and assumes that the loan exposure amount is not expected to change: • Loan exposure amount: SGD 6 million • Probability of default: 2 • Recovery rate: 95 • Volatility of the loss rate: 8% Assuming default follows a binomial distribution, what is the UL of the loan?

- A. SGD 79,825
- B. SGD 121,194
- C. SGD 264,953
- D. SGD 800,882

答案:A

解析:

A 选项正确。

•

$$UL = EA \times \sqrt{PD \times \sigma_{LR}^2 + LR^2 \times \sigma_{PD}^2}$$
$$\sigma_{PD}^2 = PD \times (1 - PD)$$

B 选项错误。

•

C 选项错误。

•

D 选项错误。

•

考察:

3.

A.

答案:B

解析:

A 选项错误。

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B 选项错误。

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C 选项错误。

•

D 选项错误。

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考察:

4.

A.

答案:B

解析:

A 选项错误。

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B 选项错误。

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C 选项错误。

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D 选项错误。

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考察:

5.

A.

答案:B

解析:

A 选项错误。

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B 选项错误。

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C 选项错误。

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D 选项错误。

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考察:

6.

A.

答案:B

解析:

A 选项错误。

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B 选项错误。

•

C 选项错误。

•

D 选项错误。

•

考察:

7.

A.

答案:B

解析:

A 选项错误。

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B 选项错误。

•

C 选项错误。

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D 选项错误。

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考察:

8.

A.

答案:B

解析:

A 选项错误。

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B 选项错误。

•

C 选项错误。

•

D 选项错误。

•

考察:

9.

A.

答案:B

解析:

A 选项错误。

•

B 选项错误。

•

C 选项错误。

•

D 选项错误。

•

考察:

4 考点 4: 监管资本

1.

A.

答案:B

解析:

A 选项错误。

•

B 选项错误。

•

C 选项错误。

•

D 选项错误。

•

考察:

2.

A.

答案:B

解析:

A 选项错误。

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B 选项错误。

•

C 选项错误。

•

D 选项错误。

•

考察:

3.

A.

答案:B

解析:

A 选项错误。

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B 选项错误。

•

C 选项错误。

•

D 选项错误。

•

考察:

4.

A.

答案:B

解析:

A 选项错误。

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B 选项错误。

•

C 选项错误。

•

D 选项错误。

•

考察:

5.

A.

答案:B

解析:

A 选项错误。

•

B 选项错误。

•

C 选项错误。

•

D 选项错误。

•

考察:

6.

A.

答案:B

解析:

A 选项错误。

•

B 选项错误。

•

C 选项错误。

•

D 选项错误。

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考察:

7.

A.

答案:B

解析:

A 选项错误。

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B 选项错误。

•

C 选项错误。

•

D 选项错误。

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考察:

8.

A.

答案:B

解析:

A 选项错误。

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B 选项错误。

•

C 选项错误。

•

D 选项错误。

•

考察:

9.

A.

答案:B

解析:

A 选项错误。

•

B 选项错误。

•

C 选项错误。

•

D 选项错误。

.

考察:

10.

A.

答案:B

解析:

A 选项错误。

.

B 选项错误。

.

C 选项错误。

.

D 选项错误。

.

考察:

11.

A.

答案:B

解析:

A 选项错误。

.

B 选项错误。

.

C 选项错误。

.

D 选项错误。

.

考察:

12.

A.

答案:B

解析:

A 选项错误。

.

B 选项错误。

.

C 选项错误。

.

D 选项错误。

.

考察: